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There are differences in early morbidity after ACL reconstruction when comparing patellar tendon and semitendinosus tendon graft

A prospective randomized study of 107 patients

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The main objective of this study was to study solely early postoperative morbidity following anterior cruciate ligament (ACL) reconstruction by comparing the gold standard procedure, the bone-patellar tendon-bone graft (BTB), and one of the most common alternatives, the semitendinosus tendon graft (ST). The prospective study included 107 randomized patients (50 BTB and 57 ST). The follow-up period was set to 20–35 weeks postoperatively (mean 26.8±3.5 weeks). One patient suffered early graft rupture and 89 (84%) of the remaining 106 patients were able to attend the follow-up within the given time limit. There were no differences in sick leave between the groups. The Lysholm score, Tegner activity level score and Visual Analog Scales (VAS) with the questions "How does your knee function?" and "How does your knee affect your activity level?" revealed no differences between the groups. Subjective patellofemoral pain, patellofemoral compartment findings and donor site morbidity were more common in the BTB group, $P<0.05$. Lachman test grade 1+ was more common in the ST group, $P<0.05$, but there was no significant difference in instrumented Lachman side-to-side comparison. The ST group scored better in the one-leg hop test than the BTB group, $P<0.05$. No correlations between these clinical and functional findings and subjective knee function scores were found. In conclusion, ACL reconstruction with ST tendon graft presented fewer short-term postoperative problems as compared to reconstruction with BTB.

Numerous studies discussing different choices of grafts in anterior cruciate ligament (ACL) reconstructive surgery have been published in recent years (Engebretsen et al., 1990; Brown et al., 1993; Aglietti et al., 1994; Anderson et al., 1994; Kleipool et al., 1998). Nevertheless, the bone-patellar tendon-bone (BTB) graft is still considered to be the gold standard procedure, mainly due to its biomechanical strength which allows safe early active rehabilitation without an increased risk of graft failure as well as documented good long-term results with regard to laxity and functional outcome (Noyes et al., 1984; Shelbourne & Wilckens, 1990; Steiner et al., 1994; Shelbourne & Gray 1997; Webb et al., 1998).

During the past decade semitendinosus and gracilis tendon grafts have increased in popularity as an alternative to the BTB. Possible advantages of the hamstrings graft compared to the BTB might be less frequent postoperative patellofemoral pain, less frequent extension deficits and less affection of the quadriceps muscle and extensor mechanism (Sachs et al., 1989; Rosenberg et al., 1992; Rosenberg & Deffner 1997; Corry et al., 1999). Despite all the previous research, there are only a few prospective randomized trials comparing BTB and other graft choices (Engebretsen et al., 1990; Marder et al., 1991; O'Neill, 1996). In addition, all of these studies mainly deal with outcome following medium- or long-term follow-up. Considering the long rehabilitation period (6–12 months) these patients require, it is our belief that valid information about possible differences between different surgical procedures and the associated rehabilitation periods is a minimum requirement for every surgeon. Moreover, the sick leave in many of these patients may last for several months postoperatively. Considering the often acceptable functional status preoperatively, as far as employment is concerned, we believe that every effort should be made to present reliable data illustrating and comparing the early morbidity following these common procedures. The final outcome is of course the main concern in all surgical procedures, but patient satisfaction in the initial postoperative rehabilitation period following ACL reconstruction tends to be a forgotten issue. On reviewing the literature, we found no previous prospective randomized studies comparing early postoperative-period morbidity in semitendinosus tendon (ST) and BTB grafts.

Consequently, the aim of this study was to analyze the possible differences in early postoperative morbidity between BTB and ST grafts as far as subjective and objective factors are concerned. The study was approved by the ethics committee at Karolinska Institutet.

Material and methods

Between late 1995 and 1997, 117 patients were prospectively randomized to reconstruction either with ipsilateral BTB graft or ipsilateral quadruple ST graft. The randomization was carried out with the closed envelope method in 100 cases and the draw method in 17 cases. The reason for proceeding with randomization, after the initial 100 cases, was to compensate for the 10 patients who were randomized despite invalid inclusion criteria. Therefore we decided to continue randomization to the end of 1997, which then included another 17 cases between September and December 1997. Inclusion criteria were unilateral anterior cruciate ligament insufficiency with the trauma occurring at least 2 months prior to reconstruction, age between 15 and 45 years, no previous cruciate ligament reconstruction in the involved knee and no concomitant posterior cruciate ligament insufficiency.

A total of 107 patients, 69 males and 38 females, fulfilled these criteria and were included in the study; 50 knees were reconstructed with BTB and 57 with ST. The mean age at the ACL reconstruction (index operation) was 27 (±6) years, mean weight 74 (±13) kg, mean height 175 (±8) cm and the mean body mass index (BMI) 24 (±3). The cause of trauma was sports-related in 95/107 (89%) of the patients. The time between the ACL trauma and the index operation was 19.8 (±28) months, with a median of 10 (3–173) months. The right knee was involved in 69 cases and the left knee in 38.

Prior to surgery, the patients were graded according to the Lysholm score, the Tegner activity level scale (present level and the level prior to the trauma) (Tegner & Lysholm, 1985) and VAS (visual analog scale), reflecting the patient's subjective opinion of "how does your knee function?" and "how does your knee affect your level of activity?" A clinical examination of the range of motion (ROM) was made and the Lachman test and pivot shift were also performed as well as an assessment of preoperative compartment findings. The donor site area was examined with inspection and palpation in order to detect any preoperative pathology that could affect the postoperative findings.

The Stryker® sagittal side-to-side laxity difference was determined at a 20-lb load (OSI Stryker, Kalamazoo, MI, USA). The one-leg hop test was performed and the thigh circumferential area 15 cm proximal to the knee joint line was measured and the difference between the nonoperated and operated leg was calculated. The operation time for each operation was noted. The cause of injury was recorded. Meniscal lesions found in the primary arthroscopic investigation following the ACL injury were recorded and treated accordingly and, when needed, additional meniscus surgery was performed simultaneously with the ligament reconstruction. Four experienced arthroscopic knee surgeons, who were familiar with both methods, performed the anterior cruciate ligament reconstructions (index operation).

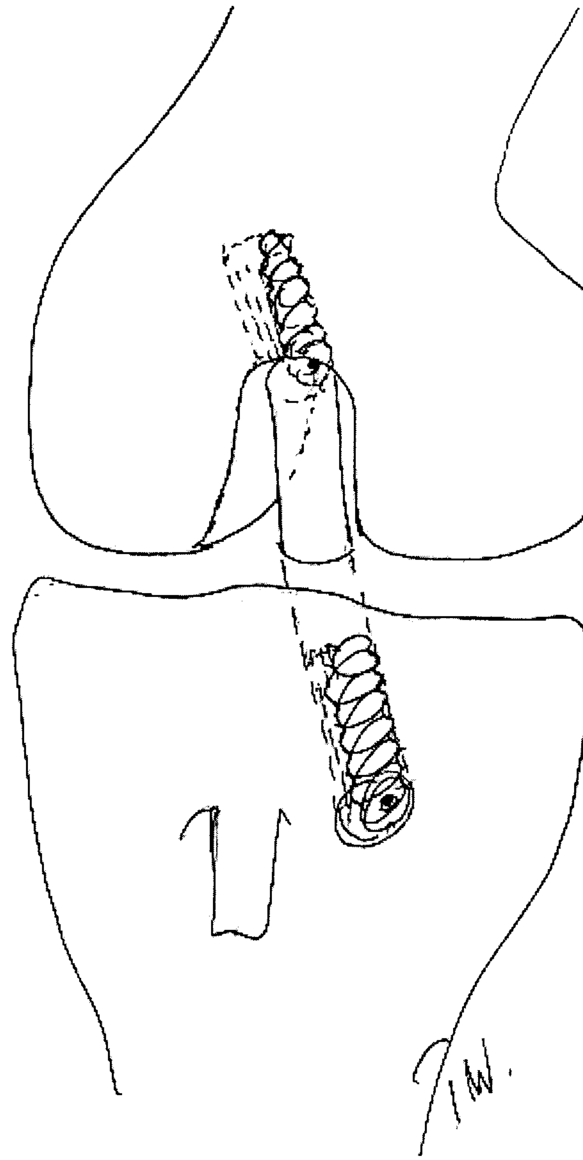

Fig. 1. Schematic illustration of ACL reconstruction with bone-patellar tendon-bone graft (BTB). The graft is fixed with interference screws proximally and distally.

Surgical technique, bone-patellar tendon-bone graft (Fig. 1)

The graft was taken through a 5 to 7-cm longitudinal skin incision over the patellar tendon from the distal part of the patella to the proximal part of the tibial tuberosity. The paratenon was incised longitudinally and the central third of the patellar tendon, along with 2.5-cm-long bone plugs from both ends, was harvested. No bone graft was used in the cortical defects. Only the paratenon was closed with absorbable sutures. The reconstruction was performed arthroscopically with one incision technique in all patients using Acufex® (Acufex Microsurgical Inc., Mansfield, MA, USA) tibial and femoral guides for tunnel placement after the remnants of the old anterior cruciate ligament had been removed. Notch plasty and additional meniscus surgery were performed when needed. The graft was fixed on the femoral side with a 7×20-mm titanium interference screw. After manual tensioning of the graft and at least ten cyclic loads with full range of motion the graft was fixed distally in full extension with a 9×20-mm titanium interference screw.

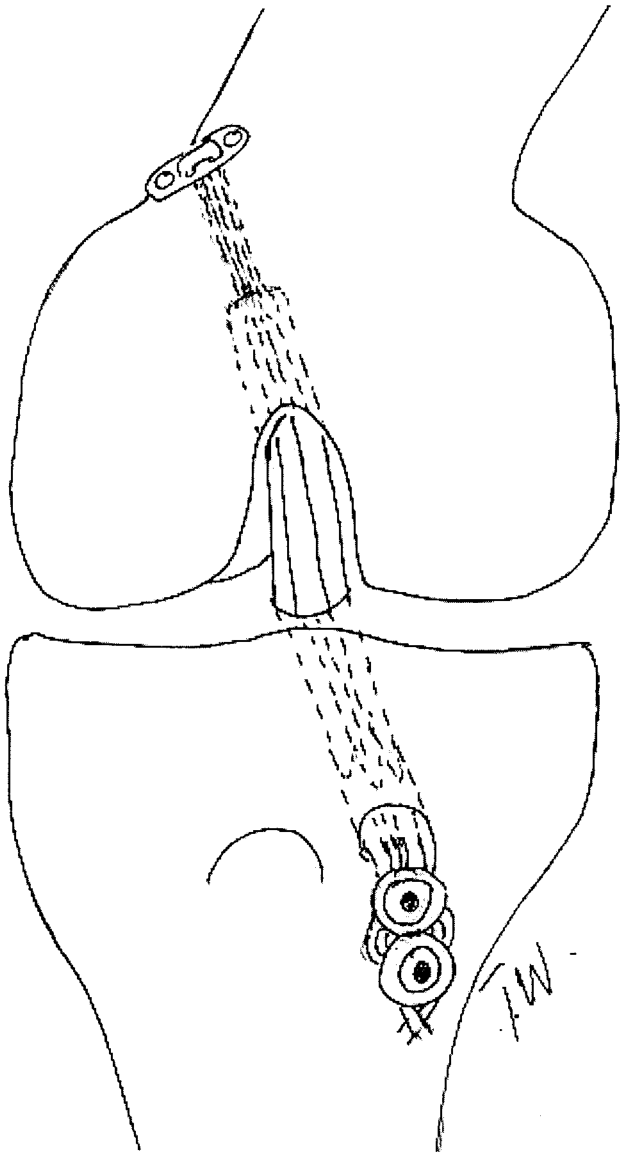

Fig. 2. Schematic illustration of ACL reconstruction with quadruple semitendinosus tendon graft (ST). The graft is fixed with endobutton proximally and around a post (2 screws and washers) distally.

Surgical technique semitendinosus tendon graft (Fig. 2)

The semitendinosus tendon was harvested through an approximately 5-cm-longitudinal skin incision over the pes anserinus, followed by an L-shaped incision in the sartorius fascia with one incision parallel to the fascia fibers. The tendon was harvested with a semiblunt, semicircular open tendon stripper (Acufex Microsurgical Inc.) after the vinculae had been dissected and cut under direct visual control. The design of the stripper ensures that the length of the graft is maximal since the harvester breaks the fibers of the semitendinosus fascia by blunt force. The length of each tendon ranged between 27 and 32 cm. When closing the wound, the sartorius fascia was meticulously closed. The tendon was prepared to a quadruple graft with mersilene tapes in each loop (two proximally and one distally), and ethibond #2 sutures, fixed as whip stitches at both free ends. The four strands were interlaced with vicryl #5.0 sutures. The median graft length was 7.5 cm (mean 7.3 cm, range 6.5–8) and the median diameter was 9 (range 8–11) mm in diameter. The graft was fixed with Endobutton® (Acufex Microsurgical Inc.) proximally and with screws distally, as described by Rosenberg (1992) and Barrett et al. (1995). The endobutton was tied at a distance from the proximal end of the graft allowing a minimum of 1.5 cm of graft material in the femoral tunnel. The graft was pretensioned by at least ten cyclic loads with a full range of motion. The distal fixation was done with the leg in full extension, in which condition the mersilene tape and ethibond sutures were tied under continuous manual tensioning of the graft, around two cortical screws with washers just distal to the tibial tunnel. The bone tunnels, intra-articular preparation and the need for notch plasty were addressed in the same manner as described for the BTB.

Postoperative management

Cryotherapy (Aircast®, Aircast Inc., Summit, NJ, USA) was administered immediately postoperatively and given for at least the first 24 h. An unlocked Genu Syncro® brace (Syncro Med. GmbH, Linz, Austria), allowing a range of motion of 0–130°, was applied for walking and was retained for three weeks. Full weightbearing was allowed, but assisted by crutches for at least three weeks.

The same rehabilitation protocol was used for all patients, including closed-chain exercises and range-of-motion training from the first postoperative day, initially under the supervision of a physical therapist. All physical therapy was administered without a brace. Open-chain exercises were allowed after 12 weeks, provided that no swelling was present and an approximate full range of motion had been regained. Running was allowed no earlier than after 12 weeks postoperatively and a return to contact and cutting sports no earlier than six months postoperatively provided that the one-leg hop test exceeded 90% of the result with the contralateral leg and that the knee was considered functionally stable.

Follow-up examination

The evaluation period was set to between 20 and 35 weeks. Patients not attending the follow-up examination within this time interval were excluded. The mean follow-up time was 26.8 (±3.5) weeks, with no significant difference between BTB and ST. The evaluations were performed by one of the surgeons. The same parameters as those assessed preoperatively were evaluated. In addition, a subjective evaluation of present patellofemoral pain, to any extent, was noted and graded: 0=no patellofemoral pain and 1=complaints of patellofemoral pain. Time for sick leave from work following the index operation was recorded for each patient. The physical demand of each occupation was graded 0–2: 0 for students, 1 for no physical demand (e.g. office worker, light lifting, salesman) and 2 for moderate or high physical demands (e.g. nurse, warehouse worker, construction worker).

Statistical analysis

Data have been summarized using descriptive statistics (mean, median and SD). The problem of multiplicity was discussed. Due to the exploratory nature of this interim analysis, any statistical significance was followed by a discussion of the clinical significance. No method for dealing with missing values was used.

Different statistical tests were performed, depending on the distribution of the variables. The ordinal variables were examined by χ^2-tests. Fisher's exact test was used if the expected cell frequency was under six. The Tegner, Lysholm and VAS scales were examined by means of non-parametric tests. The

Table 1. Preoperative clinical findings

	BTB (n=50)	ST (n=57)	Level of significance
Extension deficit (°)			
<3	48	55	ns
3–5	2	2	
Flexion deficit (°)			
0–5	48	55	ns
6–15	2	2	
Lachman test			
+1 (3–5 mm)	3	9	ns
+2 (6–10 mm)	34	34	
+3 (>10 mm)	13	14	
Pivot shift			
0	1	0	ns
+1	13	21	
+2	34	34	
+3	2	2	
Stryker laxity test, side-to-side Δ, mm 20 lb			
Mean (SD)	3.8 (1.4)	3.9 (1.8)	ns
Stryker laxity test, side-to-side Δ, mm 40 lb			
Mean (SD)	6.0 (2.0)	6.3 (2.6)	ns
Thigh circumference, side-to-side Δ (cm), 15 cm proximal to the joint line			
Mean (SD)	1.3 (1.3)	1.2 (1.2)	ns
One-leg hop test, % of contralateral side			
	n=31	n=32	
<90	10	13	ns
89–76	13	7	
75–70	7	8	
<50	1	4	
Patellofemoral compartment findings			
no crepitations (0)	44	46	ns
slight crepitations (1)	3	9	
crepitations with slight pain (2)	3	1	
crepitations with moderate pain (3)	0	1	

Wilcoxon rank test was used for differences over time. The Mann-Whitney U-test was used to analyze differences between the two surgical procedures. All tests were considered to be two sided and a significance level of 5% was employed. Spearman's rank correlation test was used to test the correlation between the parameters, and a Rho factor of >0.5 was considered to be a good correlation.

Results

There were no differences in baseline data (age, height, weight, BMI) between the BTB and ST groups. There were no statistically significant differences preoperatively between the BTB and ST groups in any of the scores, including the Lysholm subscores. Nor were there any differences between the BTB and ST groups in the preoperative clinical data (stability, ROM, compartment findings), one-leg hop and thigh circumference (Table 1). The time between the ACL trauma and the index operation did not differ between the groups. All patients had an arthroscopy performed following their trauma, i.e. prior to the index operation. At this arthroscopy 56/107 patients (52%) had meniscus injuries (37 lateral and 33 medial). 8 of the meniscus injuries, were considered stable with no need for surgical treatment, 13 (7 medial and 6 lateral) were sutured and 49 menisci were partially resected. At the time of the index operation all sutured menisci on the lateral side were healed but only 4 out of 7 on the medial side had healed, and thus 3 out of 7 were partially resected. When these 3 unhealed medial menisci were excluded, 31 patients (29%) still presented with new meniscus injuries (20 medial and 13 lateral) that had occurred between the primary arthroscopy and the index operation and thus requiring surgical treatment (partial resection; n=26, suture; n=7) at the index operation. Of the 7 sutured menisci, there were 3 in the ST and 4 in the BTB group, respectively. These patients were allowed immediate weightbearing and ROM training but no squatting for 4 weeks. There were no differences in the frequency of meniscus lesions between BTB and ST. The mean operating time was 103 (±30) min and 116 (±24) min for BTB and ST respectively, which is a significant difference, $P<0.05$.

One patient in the BTB group sustained a graft rupture while slipping on wet ground 2 months postoperatively. Of the remaining 106 patients, 89 (84%) were followed up within the specified time interval.

Table 2. Tegner activity level score for BTB and ST groups. Values are medians (range) for the whole group and percentages in each category of activity level

Activity level	BTB% (n=42)	ST% (n=47)	Level of significance
Preoperatively			
0–3	76	68	ns
4–6	24	32	
7–10	0	0	
Median (range)	2 (1–6)	2 (1–6)	
Postoperatively, 6 months			
0–3	41	38	ns
4–6	52	47	
7–10	7	15	
Median (range)	4 (2–10)	4 (2–10)	
Level of significance, pre-postoperative difference	$P<0.05$	$P<0.05$	

Table 3. Lysholm score for BTB and ST groups. Values are medians (range)

Lysholm score	BTB (n=42)	ST (n=47)	Level of significance
Preoperatively	69 (44–90)	70 (29–87)	ns
Postoperatively, 6 months	92 (64–100)	94 (62–100)	ns
Level of significance, pre-postoperative difference	$P<0.05$	$P<0.05$	

Table 4. VAS values for BTB and ST groups. Values are median (range)

VAS value	BTB	ST	Level of significance
VAS 1, preoperatively	33 (0–93) n=38	27 (0–90) n=45	ns
VAS 1, postoperatively, 6 months	82 (16–100) n=41	82 (3–100) n=47	ns
Level of significance, VAS 1 pre-postoperative difference	P<0.05	P<0.05	
VAS 2, preoperatively	20 (0–80) n=38	23 (0–99) n=45	ns
VAS 2, postoperatively, 6 months	72 (1–100) n=41	69 (14–100) n=47	ns
Level of significance, VAS 2 pre-postoperatively	P<0.05	P<0.05	

VAS 1: How does your knee function? (100=normal, 0=worst possible).
VAS 2: How does your knee affect your level of activity? (100 =normal, 0=worst possible).

For personal reasons (e.g. traveling, studies at another location, pregnancy, lack of time at the first appointment booked), 17 patients were unable to attend the clinic within the specified interval (20–35 weeks postoperatively) and were therefore excluded from further evaluation.

There were 2 early (within 14 days postoperatively) deep infections in the ST group. Neither of these patients presented positive cultures but clinical findings strongly suggested septic arthritis and they were successfully treated with arthroscopic lavage and antibiotics. In both cases 7 days of hospitalization was required. During this period no weightbearing was allowed but once these 2 patients were dismissed, weightbearing as tolerated was allowed immediately. At the follow-up these patients scored 79 and 96 respectively on the Lysholm score. One patient in the BTB group sustained a blowout fracture of the posterior femoral tunnel wall intraoperatively. This was not noted intraoperatively by the surgeon but was detected on the postoperative radiograph. The knee was considered stable intraoperatively, however, and follow-up radiographs revealed no migration of the bone block in the femoral tunnel. The rehabilitation of this patient was decelerated, however, with partial weightbearing and only passive ROM 0–90° training the first month. The Lysholm score at follow-up was 96.

The mean 100% sick leave for the whole working group was 10.3 (±6.2, n=81) weeks with no statistically significant difference between the two groups. A total of 26 patients were students and did not report any sick leave. The mean female sick leave (8.1±5.4 weeks) was significantly shorter than the mean male sick leave (11.3±6.4 weeks) ($P<0.05$). For patients with physically demanding occupations the mean period of sick leave from work was 12.9 (±6.4) weeks, and for patients with low-level physically demanding occupations the corresponding figures were 7.3 (±4.6) weeks, a significant difference ($P<0.05$).

The preoperative and postoperative Tegner, Lysholm and VAS values are shown in Tables 2, 3 and 4. Both groups improved their scores compared to those before the operation ($P<0.05$). No significant differences in these scores were found between the BTB and ST groups. Clinical findings for ROM and stability are presented in Table 5. No differences were found except a higher number of Lachman +1 (3–5

Table 5. Range of motion and stability 6 months postoperatively for BTB and ST groups

	BTB (n=42)	ST (n=47)	Level of significance
Extension deficit			
<3°	40	46	ns
3–5°	2	1	
Flexion deficit			
0–5°	37	37	ns
6–10°	5	10	
Lachman			
0 (0–2 mm)	36	28	P<0.05
+1 (3–5 mm)	6	19	
Pivot shift			
0	40	38	ns
+1 (trace)	2	9	
Stryker laxity test, side-to-side Δ, 20 lb			
Mean (SD)	0.95 (0.91)	1.1 (0.95)	ns

Table 6. Thigh circumferential difference and one-leg hop test 6 months postoperatively for BTB and ST groups

	BTB	ST	Level of significance
Thigh circumference 15 cm proximal to the joint line, side-to-side Δ, (cm)	n=40	n=45	
Mean (SD)	1.5 (1.2)	1.4 (1.3)	ns
One-leg hop	n=28	n=34	
% of nonoperated side. 27 cases missing			
≥90%	6	21	
89–76%	18	8	
75–50%	3	4	
<50%	1	1	P<0.01

Table 7. Patellofemoral compartment findings and donor site morbidity 6 months postoperatively for BTB and ST groups

	BTB	ST	Level of significance
Patellofemoral compartment findings	n=42	n=47	
no pathology=0	17	40	
crepitations=1	19	7	
crepitations and pain=2	6	0	P<0.0001
Donor site morbidity	n=42	n=47	
no=0	8	28	
light=1	20	16	
moderate/severe=2	14	3	P<0.0001

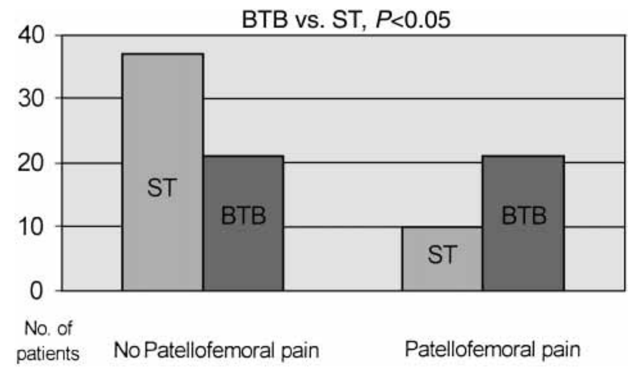

Fig. 3. Patellofemoral pain at 6-month follow-up, BTB vs. ST.

mm) in the ST group. However, the Stryker side-to-side difference at 20 lb was not significantly different. In the one-leg-hop test, a significant difference was found in favor of ST (P<0.05) (Table 6). In all, 13 cases in the ST and 14 in the BTB group chose not to perform the one-leg-hop test due to a subjective feeling of insecurity with their operated knees. No differences were found in thigh circumference between the groups (Table 6). No differences were found between BTB and ST in the medial and lateral clinical compartment findings (crepitus and/or pain). However, patellofemoral crepitus and pain was more common in the BTB group than in the ST group (P< 0.0001) (Table 7). Donor site morbidity (tenderness by palpation of the harvest site) was more common in the BTB group (Table 7). Current patellofemoral pain was more common in the BTB group (Fig. 3).

No correlations were found between functional scores (VAS, Lysholm, Tegner) and clinical variables (stability, ROM, compartment findings, donor site pathology). Nor were there any correlations between functional scores and subjective patellofemoral pain or the one-leg hop.

Discussion

The strength of this study is the prospective randomized design with a large number of patients and several surgeons involved. In fact, only a few previous prospective randomized trials comparing BTB with hamstring tendon grafts have been published, and no previous study specifically addresses the possible differences in the early postoperative phase (Marder et al., 1991; Aglietti et al., 1994; O'Neill, 1996). The weakness of the study is the lack of a validated patellofemoral knee score and the fact that one of the surgeons who had done the operation also performed the follow-up. The 84% follow-up frequency is explained by the restricted follow-up interval for inclusion in the study (20–35 weeks). The difference in operation time was significant despite the fact that the mean time difference was only 13 min. It is well known that the preparation of a quadruple semitendinosus graft takes longer than preparing the BTB graft. In a patient series with ACL reconstructions utilizing BTB, Risberg et al. (1999) found the mean VAS value for patient satisfaction 6 months postoperatively to be 73/100, which is similar to our results in both the BTB and the ST group. Our Lysholm scores, 92 and 94 for BTB and ST respectively, are high and comparable to those in previous studies on BTB at 1-, 2- and 3-year follow-up (Engebretsen et al., 1990; Otero & Hutcheson 1993; Webb et al., 1998; Kartus et al., 1999). This indicates that both methods provide equal functioning of the knee in the early phase but says nothing about future outcomes. The Tegner activity level was low (median 4) in both groups and only 7% in the BTB group and 15% in the ST group had returned to strenuous activities (Tegner 7–10) at follow-up, with no significant difference between the groups. This is within the intention of our rehabilitation program.

Patellofemoral pain, patellofemoral crepitus and donor site pathology were more common in the BTB group. However, we found no correlation between these clinical findings and functional scores, including VAS, suggesting that disability in this early phase is hard to relate to specific clinical findings. In contrast to the study by Risberg et al. (1999), range of motion was not a significant outcome measurement in our study since it did not correlate with functional scores in either the BTB or the ST group. We chose not to use the IKDC score in this study since our aim was not to compare outcomes between the methods but rather to assess possible early differences in patient satisfaction and clinical findings between BTB and ST. IKDC also has low sensitivity in detecting changes over time, according to Risberg et al. (1999). We found significant differences in favor of ST when looking at subjective patellofemoral pain and donor site pathology. Our differences in patellofemoral findings between the groups support previous outcome studies by Aglietti et al. (1994), Muneta et al. (1998) and Corry et al. (1999), in which patellofemoral problems were reported in a higher frequency with BTB than with hamstring grafts. However, no previous studies have emphasized early postoperative problems and related them to patient satisfaction. It is a well known fact that loss of motion is correlated with patellofemoral problems (Sachs et al., 1989; Aglietti et al., 1993; Shelbourne & Trumper, 1997; Kartus et al., 1999). Very few patients in our study presented extension or flexion deficits, suggesting that, in most cases, patellofemoral problems and donor site pathology without a ROM deficit are probably relatively benign findings that do not affect the functional ratings (VAS) in the early postoperative period.

There was a significant difference ($P<0.05$) in manual laxity with more Lachman 1+ values in the ST group. No significant differences were found, however, in pivot shift and Stryker side-to-side laxity at 20 lb. In accordance with Sernert et al. (1999), no correlation was found between laxity and functional scores. Steiner et al. (1994) found that gracilis-semitendinosus grafts were significantly less stiff than both the intact anterior cruciate ligament and the BTB graft in cadaver knees. This relatively lower degree of stiffness in the graft might explain the examiners' subjective Lachman interpretation with a slightly increased anterior displacement in the ST patients, although not verified in the Stryker test. In our opinion the Lachman test is a subjective evaluation of laxity, depending both on translation and stiffness of the joint, whilst the Stryker test is more objective and related to the translation only.

Thigh circumference (side-to-side difference) revealed no differences between the groups, but thigh muscle atrophy was indicated in both groups.

One interesting difference between the groups was in the one-leg hop test, where patients with knees reconstructed with ST performed better than those with knees reconstructed with BTB ($P<0.01$). To our knowledge, this has not been previously reported. The 27 patients not performing the test were evenly distributed between the groups and thus most likely without influence on the result. Our assumption is that quadriceps strength at this early postoperative phase is lower in the BTB group than in the ST group. Wilk et al. (1994) and Sekiya et al. (1998) have previously presented positive correlations between the hop test and isokinetic quadriceps strength. Aglietti et al. (1994), O'Neill (1996) and Muneta et al. (1998) all reported approximately equal isokinetic values for BTB and hamstring tendon grafts at minimum 2-year follow-ups, and possible early differences have probably disappeared over time. Another possible explanation could be preservation of proprioception in the ST group. Thus, ST might be recommended in patients requiring maximal quadriceps function for selective athletic performances. In the present study, however, we could not confirm previous findings by Wilk et al. (1994), who found a slight correlation ($r=0.48$) between one-leg hop and subjective outcome in their 6-month follow-up.

The duration of the sick-leave period did not differ between the BTB and ST tendon groups. As expected, the female group was absent from work for a shorter period than the males, probably due to less physically demanding occupations. Considering that approximately 3000 ACL reconstructions are performed in Sweden every year (senior author's judgment) and according to our series, in which two-thirds of the patients belong to the working population, the approximate total sick-leave time would be 385 working years annually in Sweden. Thus, even small improvements in the early postoperative period would have great socioeconomic implications. With any of the common surgical protocols used today (ipsilateral BTB or ST), it seems unlikely, however, that great improvements in early patient satisfaction and earlier working ability could be achieved. Considering our present findings of low donor site morbidity and previous findings showing regeneration of the semitendinosus tendon (Eriksson et al., 1999), studies using the contralateral ST could be of great interest in attempts to minimize the overall postoperative morbidity.

Meniscal injuries were found in 52% of the patients, which is well in line with previous studies (Anderson et al., 1994; Noyes & Barber-Westin, 1997).

In summary, there is a difference in one-leg hop performance, indicating that ST affects the quadriceps muscle strength or proprioception less than BTB in the early postoperative period. Early patellofemoral problems and donor site morbidity are more frequent in the BTB group but do not seem to be limiting factors in the rehabilitation. However, at 6 months in the early postoperative phase both methods yield equal patient satisfaction and functional outcomes.

Perspective

Reducing the long-lasting postoperative morbidity following ACL reconstruction remains one of the greatest challenges in sports medicine and, from that perspective, optional surgical methods must be scientifically evaluated not only with respect to final outcome but also to patient satisfaction and possible advantages in the rehabilitation period. Our results indicate some advantages for the ST in the early phase. We believe that the theoretical background for advocating hamstring tendon grafts is attractive and that randomized studies presenting data from different parts of the rehabilitation period could contribute to a wider knowledge which could eventually lead to less morbidity and shorter rehabilitation in ACL reconstructive surgery.

Based on this background and the present results, we are in favor of using the semitendinosus tendon in ACL surgery provided that long-term results do not reveal an increase in laxity of the graft.

Key words: anterior knee pain; one-leg hop; visual analog scale; sick leave; ACL reconstruction; early morbidity; semitendinosus tendon; patellar tendon.

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